

Secure Host

Secure host is the general term for the capability of a device to protect itself and the subnet from malicious software through mechanisms such as blocking access of untrusted entities to the device configuration registers, directly (through pci_cr or pci_conf) and indirectly (through MADs).

WARNING:

- Once a hardware access key is set, the hardware can be accessed only after the correct key is provided.
- If a key is lost, please refer to [Key Loss Recovery](#).

- The hardware access in this mode is allowed only if a correct 64 bits key is provided.
- The secure host feature for ConnectX-3/ConnectX-3 Pro HCAs requires a MLNX_OFED driver installed on the machine.

Using Secure Host

Secure Host feature is supported for all NVIDIA® network adapters (listed in Group 1 and group 2). For group 1 network adapters, the user is required to generate and burn a firmware image that supports the feature (see “Generating/Burning a Firmware Supporting Secure Host” below).

For Group 2 network adapters, the feature is supported on firmware version 1x.22.1002 or newer.

Generating/Burning a Firmware Supporting Secure Host

1. Make sure you have INI and mlx files suitable for the device.

2. Generate an image using mlxburn, for example run:

```
1 # mlxburn -fw ./fw-4099-rel.mlx -conf ./secure_host.ini -wimage fw-
```

2. Burn the image on the device using flint:

```
1 # flint -d /dev/mst/mt4099_pci_cr0 -i fw-4099.secure.bin b
```

3. For changes to take effect, reboot is required.

Setting the Secure Host Key

To set the key, run:

```
1 # flint -d /dev/mst/mt4099_pci_cr0 set_key 22062011
2 Setting the HW Key - OK
3 Restoring signature - OK
```

A driver restart is required to activate the new key.

Disabling/Enabling Access to the Hardware

1. Access the hardware while hardware access is disabled:

```
1 # flint -d /dev/mst/mt4099_pci_cr0 q
2 E- Cannot open /dev/mst/mt4099_pci_cr0: HW access is disabled on the device
3 E- Run "flint -d /dev/mst/mt4099_pci_cr0 hw_access enable" in order to
```

2. Enable hardware access:

```
1 # flint -d /dev/mst/mt4099_pci_cr0 hw_access enable
2 Enter Key: *****
```

3. Disable hardware access:

```
1 # flint -d /dev/mst/mt4099_pci_cr0 hw_access disable
```

Removing the Secure Host

This section is applicable to Group 1 network adapters only.

To remove the secure host feature:

1. Make sure you have INI and MLX file suitable for the device.

- a. Remove `cr_protection_en=true` from the INI (if present)
- b. Generate the image using `mlxburn`, for example run:

```
1 # mlxburn -fw ./fw-4099-rel.mlx -conf ./unsecure_host.ini -wrimage
```

2. Burn the firmware on the device (make sure hardware access is enabled prior to burning):

```
1 # flint -d /dev/mst/mt4099_pci_cr0 -i fw-4099.unsecure.bin b
```

3. Execute a driver restart in order to load the unsecure firmware:

```
1 # service openibd restart
```

Key Loss Recovery

If a key is lost, there is no way to recover it using the tool. The only way to recover is to:

1. Connect the flash-not-present jumper on the card.
2. Reboot the machine.
3. Re-burn firmware(for Group 2 network adapters re-burn the firmware following the process in [Burning a New Device](#).)
4. Remove the flash-not-present jumper.
5. Reboot the machine
6. Re-set the hardware access key

Last updated: June 17, 2021



Previous page
[flint/mlxburn Limitations](#)

Next page

[Secure Firmware Update](#)



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